

DATA STRUCTURES

Stacks & Queues: Sorting Life with Logic

Planrs Academy | Grade 6 Computer Science

Session 1: Introduction to LIFO & FIFO

Why Do We Organize?

Imagine you are looking for your favorite blue pen to finish your homework, but your desk looks like a mountain of papers, old snacks, and broken toys. What happens? You waste time, you get frustrated, and you might never find that pen!

Think about it: How do we keep our school bags or tiffin boxes neat? Why is it easier to find things when they have a "specific spot"?

In the world of computers, we have millions of pieces of information (data) coming in every second. If the computer just threw them into a messy pile, it would freeze up! That's why we use **Data Structures**.

Defining our Tools

A **Data Structure** is a specialized format for organizing, processing, retrieving, and storing data. Think of it as a "filing system" for your brain or a computer.

Real-World Organizing

- **Lunch Boxes:** You put the rice at the bottom and the curry on top.
- **Library Shelves:** Books are kept in alphabetical order.
- **Tiffin Dabbas:** They are stacked vertically to save space.

Today, we will learn the two most famous ways to organize data: **Stacks** and **Queues**.

LIFO: Last In, First Out

A **Stack** is a linear data structure where you can only add or remove items from the **TOP**. Imagine a pile of plates at a wedding buffet in India. The caterer puts a new plate on the very top of the pile. When you go to eat, which plate do you take? The one at the top!

The LIFO Principle: Last-In, First-Out. The "Last" plate placed on the pile is the "First" one to be taken by a guest.

Everyday Stacks in India

Stacks are everywhere! Here are three classic examples you see every day:

1. **The Idli Steamer:** When your mother makes idlis, the plates are stacked on top of each other. To get to the bottom idli, you must remove the top plates first.
2. **Bangles on a Wrist:** If you are wearing five bangles, the last one you slid onto your wrist is the first one you have to slide off.
3. **Notebook Pile:** When you submit your homework notebooks on the teacher's desk, the teacher starts correcting from the one on the very top.

Push & Pop

Computer scientists use two special words for adding and removing items from a stack:

Operation	Definition	Example
PUSH	Adding a new item to the TOP of the stack.	Placing a fresh chapati on top of a stack of rotis.
POP	Removing the item from the TOP of the stack.	Taking that chapati off to eat it.

Rule to Remember: You can't "Pop" an item from the middle! You must go through the top first.

FIFO: First In, First Out

A **Queue** is different. It's like a line. Imagine you are at a Railway Ticket Counter or waiting at an Auto-Rickshaw stand in Bangalore or Mumbai. Who gets to go first? The person who got there first!

The FIFO Principle: First-In, First-Out. The first person to join the line is the first person to leave the line with their ticket.

Everyday Queues in India

Queues help keep things fair in a busy country like ours:

- **Temple Darshan:** Thousands of people wait in a "Q" to see the deity. You enter at the back and exit at the front.
- **Metro Train Boarding:** People wait in a line on the platform. The first person in the line is the first one to step into the train.
- **Pani Puri Stall:** If three friends are waiting for Pani Puri, the vendor gives the first "Puri" to the friend who arrived first.

Enqueue & Dequeue

Just like Stacks, Queues have their own "magic words":

Operation	Definition	Example
ENQUEUE	Adding an item to the BACK (Rear) of the line.	Joining the end of the line at the school canteen.
DEQUEUE	Removing an item from the FRONT of the line.	Getting your food and leaving the canteen counter.

Rule to Remember: In a Queue, the entry and exit points are different ends!

LIFO vs. FIFO

Let's look at the key differences between these two ways of sorting life:

Feature	Stack (LIFO)	Queue (FIFO)
Logic	Last-In, First-Out	First-In, First-Out
Shape	Vertical Pile	Horizontal Line
Action Point	Top only	Front and Back
Fairness	Good for quick access to new things.	Good for fairness (wait your turn).

Stack or Queue?

Scenario 1: You are stacking Chapatis on a plate for dinner. You eat the one on top.

Answer: _____

Scenario 2: A line of students waiting for the School Bus.

Answer: _____

Scenario 3: A tailor has a vertical needle stand where he keeps spools of thread. He takes the topmost spool.

Answer: _____

Scenario 4: People waiting to buy tickets for a movie at PVR.

Answer: _____

What You Learned Today

- **Data Structures** are ways to organize information so it's easy to use.
- **Stacks** use LIFO (Top-only access).
- **Queues** use FIFO (Front/Back access).
- Computers use these to manage memory and run programs smoothly!

Success Goal: Can you categorize 9 out of 10 daily tasks as a Stack or a Queue? Try it tomorrow!

Show What You Know

Before you leave, grab a small piece of paper and answer this question in one sentence:

"What is the main difference between LIFO and FIFO?"

Submit this to your teacher at the door. Great job today, computer scientist!